

Day 4 (03-11-2025)

⇒ Write a Program to calculate square of a given number

```
n = int(input())
```

```
print(n*n)
```

15 ⇒ o/p ⇒ 225

⇒ a = 5
print(a*a) ⇒ type(a)
 o/p - int
 ⇒ 25

⇒ ~~a = int(input())~~ ⇒ a = input()
~~print(a)~~ type(a)
type o/p ⇒ string
o/p -

the user take input defaultly as a string. Suppose we to make it int we have to do this.

```
a = int(input())
```

```
print(a*a)
```

6 o/p ⇒ 36

Data types -

A data is classification that tells a computer what kind of value a variable holds & what operations can be performed on what data.

(or).

A data type define the type of a data a variable can store such as numbers, text or logical values.

Single value data types - It is also called primitive data types or scalar data types.
This data type that can hold only one value at a time - not collections or multiple values like arrays or objects.

- int
- float
- complex
- boolean

Multi value data types - It is also called non-primitive or composite data types)

This data type can hold (or) store multiple values at once - unlike single value data types that hold only one.

- str
- list
- tuple
- set
- dictionary

It is also called as data structures

⇒ $a = 23.8$
type(a)
⇒ float

⇒ $a = 2 + 3j$
type(a)
o/p - Complex

⇒ $a = \text{True}$
type(a)
o/p bool

⇒ $a = 2, "pooyy", 8.6, \text{false}$
type(a)
o/p tuple.

Declaring the inputs -

1) $a = 6$
 $b = 7$
 $c = 8$
print(a, b, c)

o/p → 6 7 8

2) $x = 3; y = 4; z = 5$
print(x, y, z)

o/p → 3 4 5

3) $m, n, p = 10, 20, 30$
print(m, n, p)

o/p - 10, 20, 30

Program

Write a program to swapping the number

o/p

before swap

a=10 and b=20

after swap

a=20 and b=10

try with the 3rd variable & without 3rd variable.

⇒ # 3rd variable method

a=10

b=20

print(f" before swap a={a} and b={b}")

temp=a # 3rd variable taken as temp & store a value.

a=b # now b value to a

b=temp # at last temp value to b.

print(f" after swap a={a} and b={b}")

o/p) before swap a=10 and b=20

after swap a=20 and b=10

⇒ Without 3rd variable method ②

a=10

b=20

print(f" before swap a={a} and b={b}")

~~at last~~

a, b = b, a

print(f" after swap a={a} and b={b}")

o/p) before swap a=10 and b=20

after swap a=20 and b=10

Program

2) Write a Program to calculate area of rectangle, triangle, circle?

'''
 $aor = l \times b$

$aot = 0.5 \times \text{base} \times \text{height}$

$aoc = 3.142 \times r \times r$ '''

$l = \text{float}(\text{input}(\text{"enter a len of rec:"}))$

$b = \text{float}(\text{input}(\text{"enter a breadth of rec:"}))$

$\text{base} = \text{float}(\text{input}(\text{"enter a base of tri:"}))$

$\text{height} = \text{float}(\text{input}(\text{"enter a height of tri:"}))$

$r = \text{float}(\text{input}(\text{"enter a radius of circle:"}))$

$aor = l * b$

$\text{print}(\text{"the area of rectangle = {aor} \n the area of triangle = {0.5 * base * height} \n the area of circle = {3.142 * r * r}"})$

Taking i/p ->

enter a len of rec: 5

enter a breadth of rec: 3.5

enter a base of tri: 6

enter a height of tri: 9

enter a radius of circle: 2.9

dp) the area of rec = 17.5

the area of tri = 27.0

the area of circle = 26.424219999999995